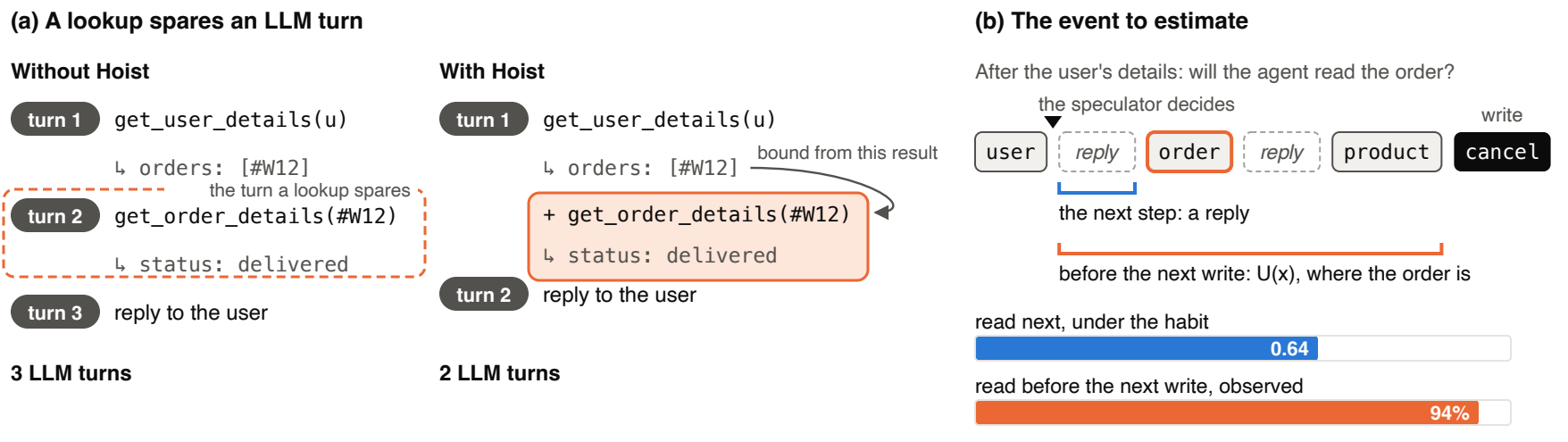

(a) A lookup spares an LLM turn
Without Hoist
turn 1
get_user_details(u)
↳ orders: [#W12]
the turn a lookup spares
turn 2
get_order_details(#W12)
↳ status: delivered
turn 3
reply to the user
3 LLM turns
With Hoist
turn 1
get_user_details(u)
↳ orders: [#W12]
bound from this result
+ get_order_details(#W12)
↳ status: delivered
turn 2
reply to the user
2 LLM turns
(b) The event to estimate
After the user's details: will the agent read the order?
the speculator decides
write
user
reply
order
reply
product
cancel
the next step: a reply
before the next write: U(x), where the order is
read next, under the habit
0.64
read before the next write, observed
94%